PROBLEM 2

Scenario 1(Normal)

● configuration 1: satisficing configuration(Using BFWS—FF-parser version solver)

Domain: ROVER_DOMAIN

Problem: ROVER_PROBLEM

Total time: 0.000739999 secs

Nodes generated during search: 255

Nodes expanded during search: 113

Plan found with cost: 31

Sequence of actions:-

Found Plan (output)

(move base sitea)

(collect-soft sitea rock1 bay2)

(move sitea base)

(deposit rock1 bay2)

(move base siteb)

(collect-soft siteb rock2 bay1)

(move siteb base)

(deposit rock2 bay1)

(move base siteb)

(move siteb sited)

(collect-soft sited rock4 bay2)

(move sited siteb)

(move siteb base)

(deposit rock4 bay2)

(move base sitea)

(move sitea sitec)

(move sitec sitee)

(pick-drill sitee)

(move sitee sitef)

(collect-hard sitef rock5 bay2)

(move sitef sitee)

(move sitee sitec)

(move sitec sitea)

(move sitea base)

(deposit rock5 bay2)

(move base sitea)

(move sitea sitec)

(collect-hard sitec rock3 bay2)

(move sitec sitea)

(move sitea base)

(deposit rock3 bay2)

Plan Validation (using VAL)

```
plan executed successfully - checking goal
plan valid
final value: 31

successful plans:
value: 31
 plan 31
```

- configuration 2: Optimal configuration(using Fast forward)

Domain: ROVER_DOMAIN

Problem: ROVER_PROBLEM

Total time: 0.0116403s

Nodes expanded during search: 1369

Plan length: 25 step(s).

Sequence of actions:-

Plan

```
(move base siteb)
(collect-soft siteb rock2 bay1)
(move siteb sited)
(collect-soft sited rock4 bay2)
(move sited siteb)
(move siteb base)
(deposit rock2 bay1)
(deposit rock4 bay2)
(move base sitea)
(move sitea sitec)
(move sitec sitee)
(pick-drill sitee)
(move sitee sitef)
(collect-hard sitef rock5 bay1)
(move sitef sitee)
(move sitee sitec)
(collect-hard sitec rock3 bay2)
(move sitec sitea)
(move sitea base)
(deposit rock5 bay1)
(deposit rock3 bay2)
(move base sitea)
(collect-soft sitea rock1 bay1)
(move sitea base)
(deposit rock1 bay1)
; cost = 25 (unit cost)
```

Plan Validation (using VAL)

```
plan executed successfully - checking goal
plan valid
final value: 25

successful plans:
value: 25
 plan 25
```

Scenario 2(Extended)

- configuration 1: satisficing configuration(Using BFWS—FF-parser version solver)

Domain: ROVER_DOMAIN

Problem: ROVER_PROBLEM_EXTENDED

Total time: 0.001449

Nodes generated during search: 409

Nodes expanded during search: 219

Plan found with cost: 60

Sequence of actions:-

Found Plan (output)

(move base sitea)
(collect-soft sitea rock1 bay1)
(move sitea base)
(deposit rock1 bay1)
(move base sitea)
(move sitea sitec)
(move sitec sitee)
(pick-drill sitee)
(move sitee sited)
(move sited siteb)
(collect-soft siteb rock2 bay2)
(move siteb sited)
(move sited sitee)

(move sitee sitec)

(move sitec sitea)

(move sitea base)

(deposit rock2 bay2)

(move base sitea)

(move sitea sitec)

(collect-hard sitec rock3 bay2)

(move sitec sitea)

(move sitea base)

(deposit rock3 bay2)

(move base sitea)

(move sitea sitec)

(move sitec sitee)

(collect-soft sitee rock6 bay2)

(move sitee sitec)

(move sitec sitea)

(move sitea base)

(deposit rock6 bay2)

(move base sitea)

(move sitea sitec)

(move sitec sited)

(collect-soft sited rock4 bay2)

(move sited sitee)

(move sitee sitec)

(move sitec sitea)

(move sitea base)

(deposit rock4 bay2)

(move base sitea)

(move sitea sitec)

(move sitec sitee)

(move sitee sitef)

(collect-hard sitef rock7 bay2)

(move sitef sitee)

(move sitee sitec)

(move sitec sitea)
(move sitea base)
(deposit rock7 bay2)
(move base sitea)
(move sitea sitec)
(move sitec sitee)
(move sitee sitef)
(collect-hard sitef rock5 bay2)
(move sitef sitee)
(move sitee sitec)
(move sitec sitea)
(move sitea base)
(deposit rock5 bay2)

Plan Validation (using VAL)

```
plan executed successfully - checking goal
plan valid
final value: 60

successful plans:
value: 60
 plan 60
```

● configuration 2: Optimal configuration(using Fast forward)

Domain: ROVER_DOMAIN

Problem: ROVER_PROBLEM_EXTENDED

Total time: 0.0861807s

Plan length: 40 step(s)

Nodes expanded during search: 14208

Sequence of actions:-

Plan

(move base sitea)
(move sitea sitec)
(move sitec sited)
(collect-soft sited rock4 bay1)
(move sited siteb)
(collect-soft siteb rock2 bay2)
(move siteb sited)
(move sited sitee)
(pick-drill sitee)
(move sitee sitec)
(move sitec sitea)
(move sitea base)
(deposit rock4 bay1)
(deposit rock2 bay2)
(move base sitea)
(move sitea sitec)
(move sitec sitee)
(move sitee sitef)
(collect-hard sitef rock5 bay1)
(collect-hard sitef rock7 bay2)
(move sitef sitee)
(move sitee sitec)
(move sitec sitea)
(move sitea base)
(deposit rock5 bay1)
(deposit rock7 bay2)
(move base sitea)
(move sitea sitec)
(collect-hard sitec rock3 bay1)
(move sitec sitee)

```
(collect-soft sitee rock6 bay2)
(move sitee sitec)
(move sitec sitea)
(move sitea base)
(deposit rock3 bay1)
(deposit rock6 bay2)
(move base sitea)
(collect-soft sitea rock1 bay1)
(move sitea base)
(deposit rock1 bay1)
; cost = 40 (unit cost)
```

Plan Validation (using VAL)

```
plan executed successfully - checking goal
plan valid
final value: 40

successful plans:
value: 40
 plan 40
```